SMART INDIA HACKATHON — SOLUTION BLUEPRINT

MediKiosk+

The AI Front Door to Every Consultation

An AI-Powered Multimodal Clinical History & Document Intelligence Platform for AYUSH & Allopathic OPDs

Field	Detail
Problem Statement ID	26047
Problem Statement Title	Patient Case-Taking Software
Organization	Ministry of Ayush
Department	All India Institute of Ayurveda (AIIA)
Category	Software
Theme	MedTech / BioTech / HealthTech

1. Understanding the Problem Statement

Problem Statement 26047, issued by the Ministry of Ayush through the All India Institute of Ayurveda, targets a structural failure at the very first step of clinical care: history-taking. The brief is precise about the pain points, and any winning solution must be judged first on how faithfully it solves exactly this, before it is judged on originality.

1.1 The Core Bottleneck

- OPD consultation time in Indian tertiary hospitals is compressed to 2–5 minutes per patient, against a global average that is already thin.
- A complete history is classically responsible for 70–80% of correct diagnoses — yet it is the first casualty of time pressure.
- AYUSH (Ayurvedic) intake is structurally heavier than allopathic intake: Trividha, Ashtavidha and Dashavidha Pariksha require Prakriti, Vikriti, Agni, Koshtha, Ahara-Vihara, Nidana and Samprapti assessment — a framework with no equivalent time allowance in a 5,000-patient-a-day OPD.
- Patients arrive with fragmented paper records — handwritten, multilingual, chronologically scattered — that consume consultation time simply to sort through.

1.2 Why the Ministry Frames This as a Software (not App) Problem

The problem statement deliberately asks for a 'software platform' rather than a mobile app, and lists ABDM (ABHA IDs, Health Information Exchange, FHIR) as the target ecosystem. This signals three hard requirements that many teams will under-weight:

1. It must work for a walk-in, first-visit, no-smartphone, low-literacy patient — not just tech-comfortable users.
2. It must plug into the ABDM 'first mile' — capturing structured history before the encounter and pushing it into the HIS/EMR and the ABHA Personal Health Record via FHIR, not existing as an island app.
3. It must serve two distinct clinical grammars simultaneously — modern allopathic ROS-based history and classical Ayurvedic Dashavidha Pariksha — inside one adaptive engine.

1.3 Gap Analysis of Existing Solutions

Existing Approach	What It Does	Why It Falls Short
Hospital registration kiosks	Captures name, age, department, token	Zero clinical history, zero document intelligence
Mobile health apps / tele-triage bots	Chat-based symptom capture	Needs a smartphone, data plan, and pre-enrolment — excludes elderly, rural, first-visit patients
Nurse-led triage desks	Human history-taking at the door	Doesn't scale past a few hundred patients/day; reproduces the same time bottleneck
Generic document scanner apps	Digitizes an image to PDF/JPEG	No OCR-to-structured-data, no timeline, no ABHA linkage

Judging Insight:
Evaluators from AIIA will be listening for how well a team understands Ayurvedic history-taking specifically — not just a generic 'AI medical chatbot' repackaged for the brief. Depth on Dashavidha Pariksha and Ahara-Vihara is a genuine differentiator most competing teams will skip.

2. Solution Vision — MediKiosk+

MediKiosk+ builds on the four modules implied by the problem statement (Conversational History, Document Intelligence, Summary Generation, Consent/ABDM) and layers a fifth capability the brief does not ask for but every judge will notice: it closes the loop after the consultation and after the hospital, not just before it.

2.1 One-Line Pitch

> **Pitch:**
> "MediKiosk+ turns the 5 minutes a doctor doesn't have into 5 minutes the patient already spent productively — talking, tapping, and scanning in their own language, before they ever sit down."

2.2 Design Principles

- Zero-training usability — a 70-year-old, first-time, non-literate patient must complete intake unaided.
- Physician stays in control — every AI output is a draft; nothing is written to the medical record without a clinician's confirm/edit.
- Dual clinical grammar — the same conversational engine speaks fluent SOCRATES (allopathic) and fluent Dashavidha Pariksha (Ayurvedic), switched by OPD type, not by rebuilding the product.
- Works where connectivity doesn't — offline-first kiosk with deferred sync, because a Ministry of Ayush deployment includes rural and semi-urban AYUSH OPDs.
- Privacy by architecture, not by policy — voice and document processing minimized and purged, aligned to DPDP Act 2023 and ABDM consent rules.

3. System Architecture

3.1 High-Level Layers

- Presentation Layer — kiosk touchscreen UI, companion web/mobile link (QR handoff), IVR fallback for feature phones.
- Interaction Layer — ASR (Indian languages), TTS, dialogue manager, avatar/sign-language layer.
- Intelligence Layer — clinical history LLM (ontology-constrained), OCR + medical NER, red-flag/triage classifier, summarization engine.
- Integration Layer — FHIR adapter, ABDM/ABHA connector, HIS/EMR push API, WhatsApp/SMS/IVR notification gateway.
- Data & Security Layer — encrypted transient session store, consent ledger, audit logging, DPDP-compliant purge jobs.

3.2 Modules Carried Forward from the Brief

Module	Function
A — Conversational Multimodal History Engine	Voice + touch adaptive interview; SOCRATES branching; AYUSH Dashavidha Pariksha mode; red-flag detection
B — Medical Document Digitization & Intelligence	Multilingual OCR (print + handwriting); entity extraction; chronological timeline; abnormal-value flagging
C — Structured History Summary Generator	Merges conversation + documents into a standard, editable, bilingual clinical summary
D — Consent, Privacy & ABDM Integration	ABHA authentication, granular consent, FHIR push to HIS/EMR, session purge

3.3 New Modules That Differentiate MediKiosk+

Module	What It Adds	Why It Wins
E — AI Queue Prioritization	Converts red-flag detection into a live, explainable acuity score that reorders the OPD queue, not just alerts staff	Turns a safety feature into an operational efficiency feature administrators can measure
F — Continuity & After-Visit Loop	Post-consultation summary + medication/follow-up reminders via WhatsApp/IVR, linked back to ABHA	Solves the 'last mile' the brief leaves open — most teams stop at check-in
G — Ayurveda Depth Pack	Structured Prakriti/Vikriti card-based self-assessment, Ahara-Vihara diary import, optional camera-assisted Nadi/tongue image capture flagged as research-only, non-diagnostic	Directly answers the AIIA-specific half of the brief that generic HealthTech teams will gloss over
H — Accessibility & Inclusion Layer	Sign-language avatar, audio-first flow, Braille-compatible tactile buttons, voice-biometric repeat-visit login	Matches the brief's own emphasis on elderly/low-literacy/rural patients with concrete features, not a slogan
I — Model Feedback Loop	Every physician edit to the AI summary is captured as a labeled correction and used to fine-tune the summarizer	Shows judges a credible plan for the system to keep improving after deployment, not just at demo time
J — Federated Analytics Dashboard	Anonymized, aggregated symptom/diagnosis trends surfaced to hospital administration and, if permitted, the Ministry	Gives the problem owner (Ministry of Ayush) a reason to want this beyond a single hospital pilot

4. Feature Set — Mandatory vs. Differentiating

4.1 Mandatory Features (must work flawlessly in the demo)

4. Multilingual voice + touch history capture with adaptive follow-up questioning (SOCRATES for allopathic chief complaints).
5. AYUSH mode: full Dashavidha Pariksha interview (Prakriti, Vikriti, Sara, Samhanana, Pramana, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, Vaya) plus Ahara-Vihara capture.
6. Document upload/scan with OCR for printed and handwritten prescriptions, lab reports, discharge summaries.
7. Automatic chronological timeline of prior documents with abnormal-value highlighting.
8. AI-generated structured clinical summary (Chief Complaint → HPI → Past History → Drug/Allergy → Family → Personal → ROS → Investigations) — editable by the physician.
9. ABHA-based authentication, explicit granular consent with audio explanation, FHIR push to HIS and ABHA PHR.
10. Red-flag detection with immediate triage alert for emergency presentations.
11. Session data purge immediately after successful submission.

4.2 Differentiating / “Out-of-the-Box” Features

Beyond Check-In: Closing the Loop

- After-visit digital summary + prescription pushed to the patient's ABHA PHR and sent via WhatsApp/IVR in their language, so the same kiosk experience that saved the doctor's time also saves the patient a second trip for clarification.
- Medication and follow-up-visit reminders via IVR call for patients without smartphones — extending ABDM's reach to the actual first-visit demographic the brief describes.

Ayurveda-Native Intelligence

- Card-based, illustrated Prakriti self-assessment so a non-literate patient can identify constitutional traits by tapping pictures/audio rather than reading a questionnaire.
- Optional camera-assisted tongue/pulse-region image capture, explicitly labeled a research-data aid for the Vaidya's manual Nadi Pariksha — never presented as an automated diagnosis, keeping the physician as the sole diagnostic authority.

Operational Intelligence for Hospital Administration

- Explainable acuity scoring reorders — never auto-diagnoses — the OPD queue, with a plain-language reason shown to triage staff (e.g., “chest pain + breathlessness reported”).
- Anonymized, hospital-level analytics dashboard surfacing common presenting complaints, seasonal disease trends, and average time-saved per consultation — a tool the hospital superintendent and the Ministry can actually use to justify scale-up.

Trust, Accessibility & Continuous Improvement

- Sign-language avatar and full audio-first navigation for hearing- and vision-impaired patients.
- Voice-biometric recognition for repeat patients, cutting re-registration time on follow-up visits.
- A structured feedback loop where every physician correction to an AI summary becomes a labeled training example — presented to judges as the system's plan for getting measurably better after deployment.
- Offline-first kiosk mode: history and OCR processing continue on local compute during connectivity loss, with deferred, encrypted sync to HIS/ABDM once network is restored.

5. Technology Stack

Layer	Suggested Technology
Kiosk UI	React / Flutter (touch-first, large-target UI), offline-capable PWA shell
ASR (Indian languages)	Bhashini / AI4Bharat ASR models, fine-tuned on hospital-noise audio
TTS	Bhashini / AI4Bharat TTS for local-language audio prompts
Dialogue Manager	Ontology-constrained LLM orchestration (clinical history schema + AYUSH Dashavidha schema)
OCR & Document AI	Layout-aware multilingual OCR + medical NER for entity extraction
Summarization Engine	LLM fine-tuned on structured clinical note generation, with mandatory human-in-the-loop review
Interoperability	HL7 FHIR adapters, ABDM Health Information Exchange & Scan-and-Share, ABHA auth SDK
Backend	Microservices (Node.js/Python), event-driven pipeline for ASR → NLU → summarizer → FHIR push
Data Store	Encrypted transient session store; no long-term PHI retention outside HIS/ABDM
Notifications	WhatsApp Business API / SMS gateway / IVR for after-visit loop
Analytics	Anonymization/aggregation pipeline feeding a hospital-admin dashboard

6. End-to-End Patient & Physician Journey

Step	What Happens
1. Identify	ABHA ID / Aadhaar scan or new registration; language selection; audio-guided granular consent
2. Converse	Adaptive voice + touch interview; allopathic SOCRATES or AYUSH Dashavidha Pariksha mode; red-flags trigger priority alert
3. Scan	Prior prescriptions/labs/discharge summaries digitized, structured, timelined, abnormal values flagged
4. Summarize & Route	AI drafts structured summary; links to ABHA; pushes to HIS; queue reprioritized if a red-flag was raised
5. Consult	Physician reviews the draft in seconds, edits/confirms; correction is logged for model improvement
6. Continue	After-visit summary + reminders sent via ABHA PHR, WhatsApp, or IVR; follow-up nudge scheduled

7. Privacy, Consent & Compliance

- DPDP Act 2023 alignment: purpose limitation, data minimization, and a documented consent record per session.
- ABDM consent framework: consent artefact generated and stored per interaction, revocable by the patient at any time.
- Session-scoped processing: raw voice audio and document images are processed transiently and purged immediately after the structured summary is confirmed and pushed to HIS/ABDM.
- Role-based access: only the treating physician sees the full draft summary before confirmation; administrators see only anonymized aggregate analytics.
- Human-in-the-loop guarantee: the system is explicitly positioned as a documentation and triage-support aid — never an autonomous diagnostic tool — which is both an ethical requirement and a strong point to make to judges.

8. Hackathon Build Roadmap

A realistic plan for a 24–36 hour internal hackathon build, sequenced so that a working demo exists at every checkpoint even if later stages slip.

Phase	Timebox	Deliverable
0. Problem framing & scope lock	Hour 0–2	Finalize demo persona set (allopathic + AYUSH), lock the exact summary schema
1. Core conversational flow	Hour 2–8	Voice+touch intake for one chief complaint (e.g., chest pain) with SOCRATES branching, in 2 languages
2. Document digitization	Hour 8–14	OCR pipeline on 3–4 sample prescriptions/lab reports; entity extraction; timeline view
3. Summary generator	Hour 14–20	Merge conversation + documents into the standard editable summary screen
4. AYUSH mode + differentiators	Hour 20–26	Dashavidha Pariksha card UI; red-flag → queue reprioritization; after-visit WhatsApp/IVR mock
5. Integration polish	Hour 26–30	Mock ABHA auth + FHIR push demo, consent screen, offline-mode toggle demo
6. Rehearsal & pitch deck	Hour 30–36	End-to-end run-through, judge-facing narrative, backup video recording of the full flow

Demo Safety Net:
Record a clean end-to-end video of the full flow by the second-to-last checkpoint. Live ASR/OCR demos are the single most common hackathon failure point — always have a recorded fallback ready.

9. Why This Wins the Internal Round

- Faithful to the brief: every mandatory element in Problem Statement 26047 — dual-mode input, AYUSH history depth, document digitization, ABDM/FHIR integration, DPDP-compliant consent — is addressed explicitly, so the solution can't be marked down on scope.
- AYUSH-native, not AYUSH-adjacent: the Dashavidha Pariksha card UI and Ahara-Vihara capture show real domain understanding to an AIIA judging panel, not a generic chatbot with an Ayurveda label bolted on.
- Solves a problem the brief hints at but doesn't name: the after-visit continuity loop (Module F) demonstrates systems thinking beyond the literal ask.
- Operationally credible: the queue-prioritization and admin analytics modules give a hospital administrator — not just a patient — a reason to champion adoption, which matters in government deployment contexts.
- Ethically framed: keeping the physician as the sole diagnostic authority at every stage pre-empts the most obvious judge objection to AI-in-healthcare pitches.

10. Key Risks & Mitigations

Risk	Mitigation
ASR accuracy drops in noisy OPD environments / regional accents	Dual-mode input (voice + touch) as fallback; noise-robust ASR fine-tuning; confirm-back prompts
OCR fails on poor-quality handwritten prescriptions	Confidence-scored extraction with manual-correction UI shown to patient/staff before submission
Patients without ABHA ID	On-the-spot ABHA creation flow embedded in Step 1; graceful fallback to Aadhaar-based registration
Over-trust in AI summary by a rushed physician	Summary is never auto-saved; explicit doctor confirm/edit step is mandatory before it enters the record
Connectivity loss in rural AYUSH OPDs	Offline-first kiosk mode with local processing and deferred, encrypted sync

11. Conclusion

MediKiosk+ answers Problem Statement 26047 exactly as written — a patient-facing, AI-driven, multimodal history and document-intelligence platform integrated into ABDM — while adding the operational, continuity, and Ayurveda-depth layers that turn a compliant submission into a distinctive one. The build roadmap is scoped to produce a working, demoable system within a single hackathon window, with a recorded fallback to protect against live-demo risk.

The strongest pitch to the panel is simple: every other team will build a history-taking chatbot. This team builds the first mile of a national digital health record that also happens to give the doctor their five minutes back — and gives the patient, the hospital administrator, and the Ministry each a distinct reason to want it deployed.